Today's Vendetta

① NFAs

② Proofs of completion of reg. lang. [brief disc.]

③ DFAs are equivalent to NFAs.

④ Pumping lemma for regular lang.

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① **NFA**: defined as a 5-tuple,

$(\underbrace{Q}_{\substack{\text{finite, non-empty}\\ \text{set of states}}}, \underbrace{\Sigma}_{\substack{\text{(finite) set}\\ \text{of symbols}}}, \overbrace{\delta}^{\text{transition function}}, \underbrace{q_0}_{\text{start state}}, \underbrace{F}_{\text{accepting states}})$

$\delta : Q \times (\Sigma \cup \{\epsilon\}) \longrightarrow \mathcal{P}(Q)$ (power set of Q)

NFA examples:

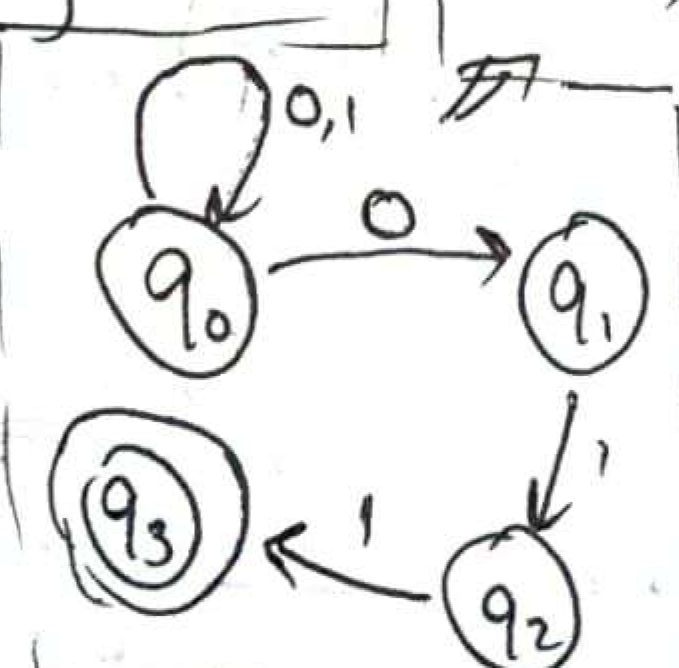
① $L := \{ w \mid w \text{ ends with } 011 \}$

$Q = \{q_0, q_1, q_2, q_3\}$

q_0 → Start state

q_3 → accepting states

δ	$P(Q)$
$(q_0, 0)$	$\{q_0, q_1\}$
$(q_0, 1)$	$\{q_0\}$
$(q_1, 1)$	$\{q_2\}$
$(q_2, 1)$	$\{q_3\}$
all other	ϕ



How to check a string of NFA? 'Active set'

$\boxed{11011}$

$\to \{q_0\} \xrightarrow{1} \{q_0\} \xrightarrow{1} \{q_0\} \xrightarrow{0} \{q_0, q_1\} \xrightarrow{1} \{q_0, q_2\} \xrightarrow{1} \boxed{\{q_0, q_3\}}$

q_3 is an accepting state, thus $11011 \in L$

$\boxed{01101}$

$\to \{q_0\} \xrightarrow{0} \{q_0, q_1\} \xrightarrow{1} \{q_0, q_2\} \xrightarrow{1} \{q_0, q_3\} \xrightarrow{0} \{q_0, q_1\} \xrightarrow{1} \boxed{\{q_0, q_2\}}$

no accepting states!

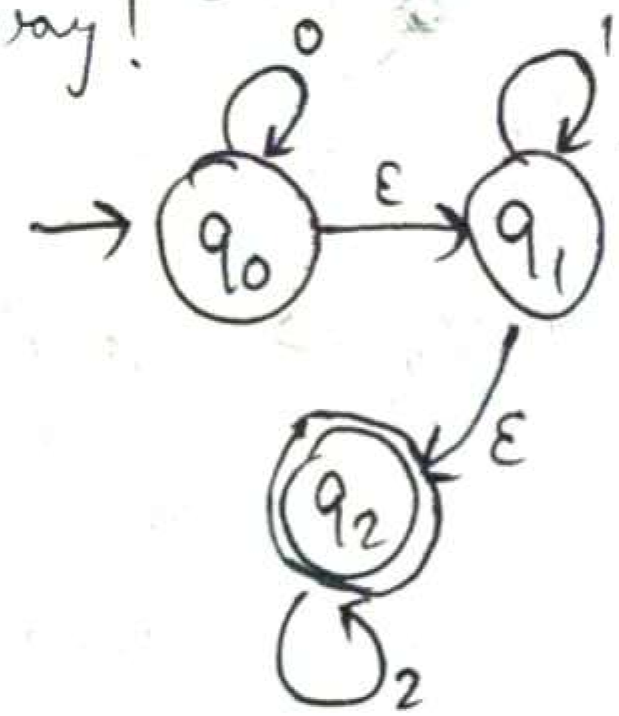
So, $01101 \notin L$

② $L = \{ w \mid w = 0^* 1^* 2^* \}$

≥ 0 many occurences of 0 ; " " " of 1 ; " " " of 2

So, $\varepsilon \in L$ by the way!

δ	$P(Q)$
$(q_0, 0)$	q_0
(q_0, ε)	q_1
$(q_1, 1)$	q_1
(q_1, ε)	q_2
$(q_2, 2)$	q_2



How does this even reject anything?

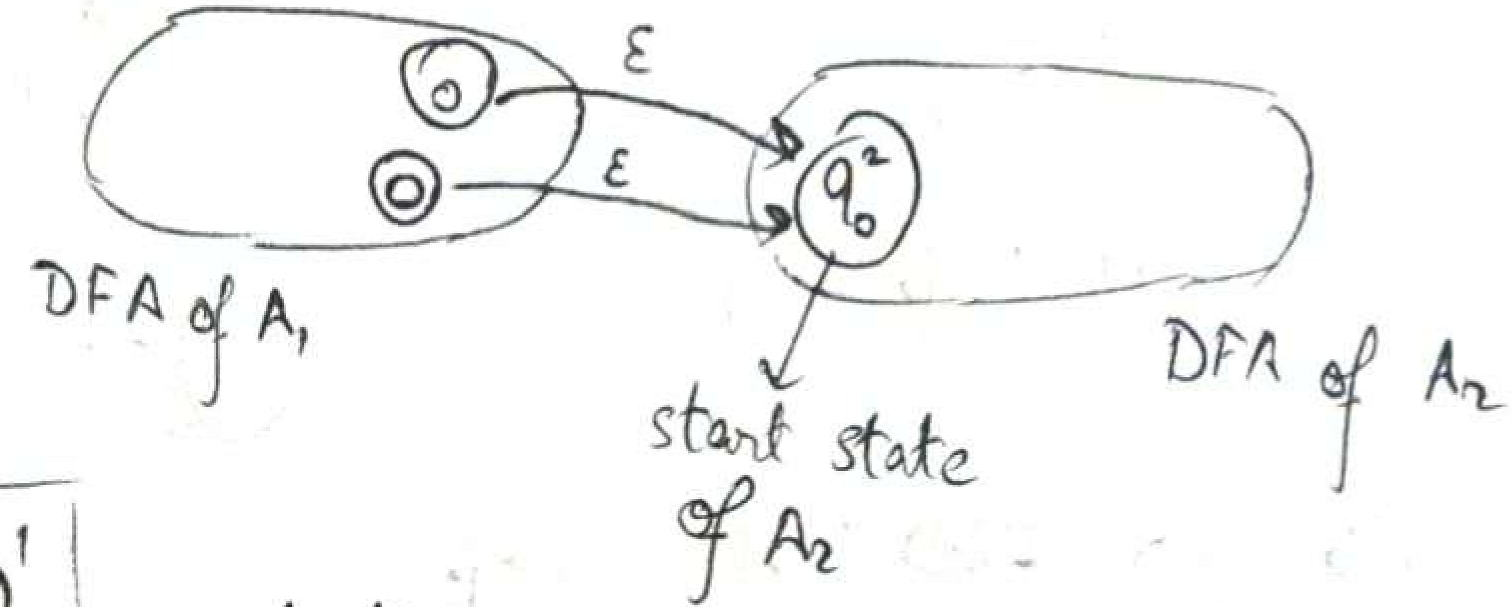
$\boxed{\varepsilon}$ Firstly, $\{q_0\} \xrightarrow{\varepsilon} \{q_0, q_1\} \xrightarrow{\varepsilon} \{q_0, q_1, q_2\}$

$\boxed{21}$ $\{q_0, q_1, q_2\} \xrightarrow{2} \{q_2\} \xrightarrow{1} \phi$

$\boxed{10}$ $\{q_0, q_1, q_2\} \xrightarrow{1} \{q_1, q_2\} \xrightarrow{\varepsilon} \{q_1, q_2\} \xrightarrow{0} \phi$

② A_1, A_2 regular languages $\Rightarrow A_1 \circ A_2$ are regular too!

↳



rest all was mostly verbal.

③ <u>Equivalence of DFA & NFA.</u>

↳ every DFA has an equivalent NFA.

<u>proof:</u> Let us consider DFA $(Q, \Sigma, \delta, q_0, F)$

Our NFA $(Q_N, \Sigma, \hat{\delta}, q_0^N, F_N)$ is defined as follows:

$Q_N = Q$; Σ is same;

$q_0^N = q_0$; $F_N = F$ &

$\hat{\delta}(q, \alpha) = \{ \delta(q, \alpha) \} \ \forall q \in Q_N, \forall \alpha \in \Sigma$

trivial <u>equivalence</u>

↳ every NFA has an equivalent DFA.

[Subset Construction Method]

① create ~~stat~~ start state

② compute transitions for each input

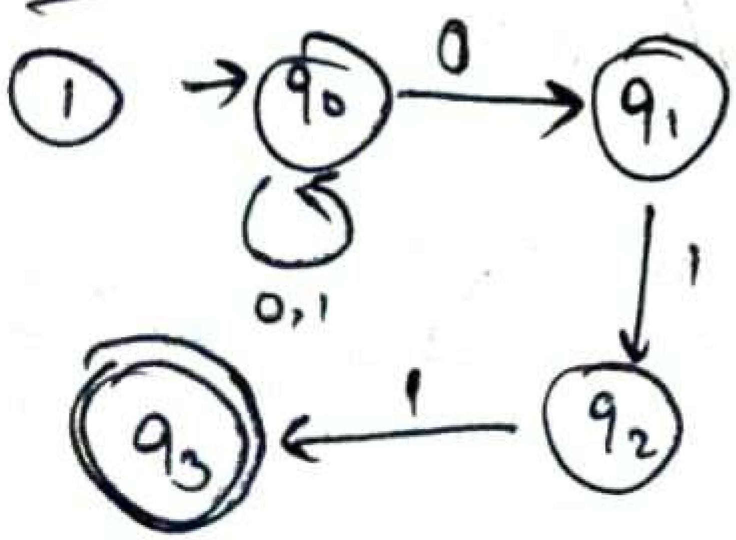
↳ every time you see a new subset of $P(Q)$, add that to list of states

③ repeat ② for new states

④ assign accept states

Examples:

①



NFA.

(i) Start with $\{q_0\}$

$\{q_0\}, 0 \longrightarrow \boxed{\{q_0, q_1\}}$ New!

$\{q_0\}, 1 \longrightarrow \{q_0\}$

$\{q_0, q_1\}, 0 \longrightarrow \{q_0, q_1\}$

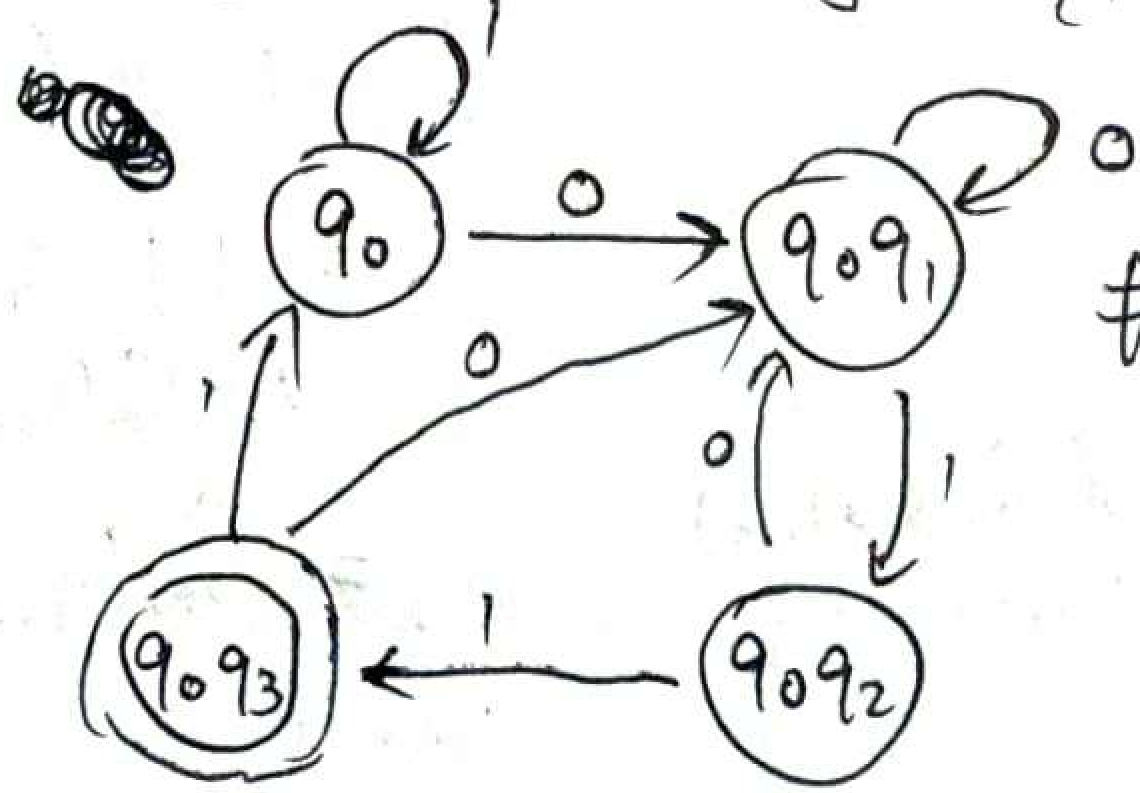
$\{q_0, q_1\}, 1 \longrightarrow \boxed{\{q_0, q_2\}}$ New!

$\{q_0, q_2\}, 0 \longrightarrow \{q_0, q_1\}$

$\{q_0, q_2\}, 1 \longrightarrow \boxed{\{q_0, q_3\}}$ New!

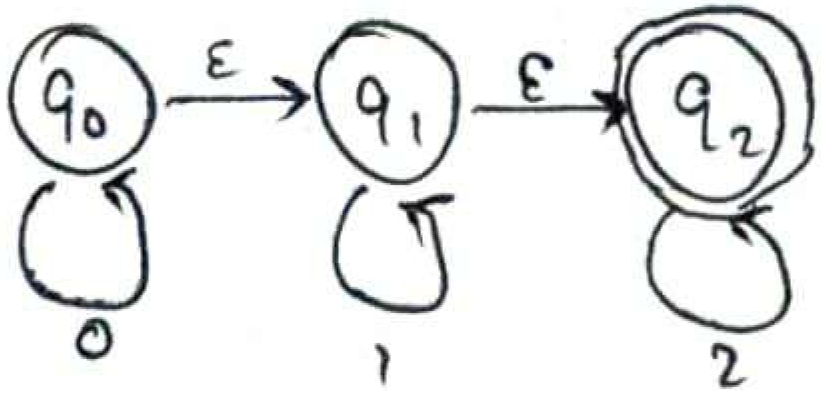
$\{q_0, q_3\}, 0 \longrightarrow \{q_0, q_1\}$

$\{q_0, q_3\}, 1 \longrightarrow \{q_0\}$



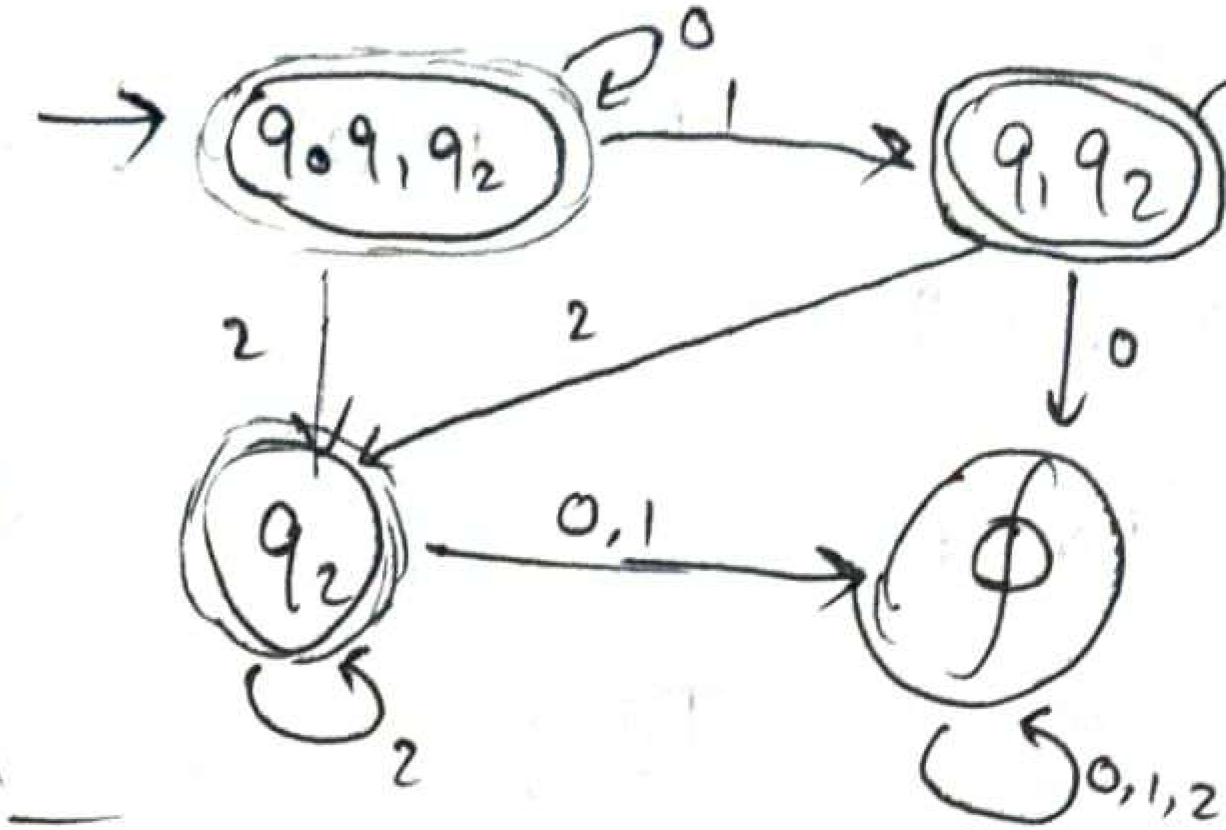
q_0q_3 is acc. state as q_3 is in it.

② $\{w \mid w = 0^*1^*2^*\}$



Due to ε transitions,

$\{q_0, q_1, q_2\}$ is our start state.



$\boxed{\text{DFA}}$

$\{q_0, q_1, q_2\} \xrightarrow{0} \{q_0, q_1, q_2\}$

$\quad \xrightarrow{1} \boxed{\{q_1, q_2\}}$ New!

$\quad \xrightarrow{2} \boxed{\{q_2\}}$ New!

$\{q_1, q_2\} \xrightarrow{0} \boxed{\phi}$ New!

$\quad \xrightarrow{1} \{q_1, q_2\}$

$\quad \xrightarrow{2} \{q_2\}$

$\{q_2\} \xrightarrow{0,1} \phi$

$\quad \xrightarrow{2} \{q_2\}$

$\phi \circlearrowleft 0,1,2$

<u>Proof for NFA has equiv. DFA:</u>

$N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ takes in lang. L.

WLOG, let us <u>not</u> have any ε-transitions

Note: because, if we take ε-transitions, just plug in $E(q)$ instead of q, where $E(q)$ is set of states to which q has ε-transitions to.

$q \xrightarrow{\varepsilon} l$ either you have only $\{l\}$ or $E(q) := \{q, l\}$

$Q_D = P(Q_N)$

$q_D = \{q_0\}$

Accepted states of DFA (F_D)

$= \{ R \in Q_D \mid R \cap F_N \neq \phi \}$

Transition function

$$\delta_D : Q_D \times \Sigma \longrightarrow Q_D$$

$$\delta_D(R, a) = \bigcup_{r \in R} \delta_N(r, a)$$

Now, <u>need</u> to show $\boxed{L(N) = L(D)}$

* <u>Extended transition function</u>:

$\hat{\delta}$ handles a <u>string</u> of characters & is defn. recursively.

for DFA,

- $\hat{\delta}_D(R, \varepsilon) = R$ [base case]
- $\hat{\delta}_D(R, xa) = \delta_D(\hat{\delta}_D(R, x), a)$ [recursive]

for NFA,

- $\hat{\delta}_N(q, \varepsilon) = \{q\}$ [base]
- $\hat{\delta}_N(q, xa) = \bigcup_{r \in \hat{\delta}_N(q, x)} (\delta_N(r, a))$ [recursive]

Let us take $|w| = 0$.

$\Rightarrow w = \varepsilon$.

$\hat{\delta}_D(\{q_0\}, \varepsilon) := \{q_0\} =: \hat{\delta}_N(q_0, \varepsilon)$

↳ holds for $|w| = 0$

Assume lemma holds for some arbitrary string x of length k.

$\Rightarrow \hat{\delta}_D(\{q_0\}, x) = \hat{\delta}_N(q_0, x) = S$. (say)

Let $w = xa$, $a \in \Sigma$ [for some a]

$$\hat{\delta}_D(\{q_0\}, xa) = \delta_D(\hat{\delta}_D(\{q_0\}, x), a)$$

$$= \delta_D(S, a) = \bigcup_{r \in S} \delta_N(r, a)$$

$$= \bigcup_{r \in \hat{\delta}_N(q_0, x)} \delta_N(r, a) \quad \text{[by defn of } S\text{]}$$

$$= \hat{\delta}_N(q_0, xa)$$

Now, a string w is accepted by D $\Leftrightarrow$ state it ends in is in F_D

So, $w \in L(D) \Leftrightarrow \hat{\delta}_D(\{q_0\}, w) \in F_D$

$\Leftrightarrow \hat{\delta}_D(\{q_0\}, w) \cap F_N \neq \phi$

$\Leftrightarrow \hat{\delta}_N(\{q_0\}, w) \cap F_N \neq \phi$

$\Leftrightarrow$ the string w is accepted by accepted by N

④ Pumping Lemma for Regular Lang.

↳ test for seeing if lang is regular.

Statement:

If L is regular, then

$\exists$ a constant $p \geq 1$, such that,

any string $w \in L$ with length $\geq p$, can be split into three substrings, $w = xyz$, satisfying the following conditions:

① $|xy| \leq p$

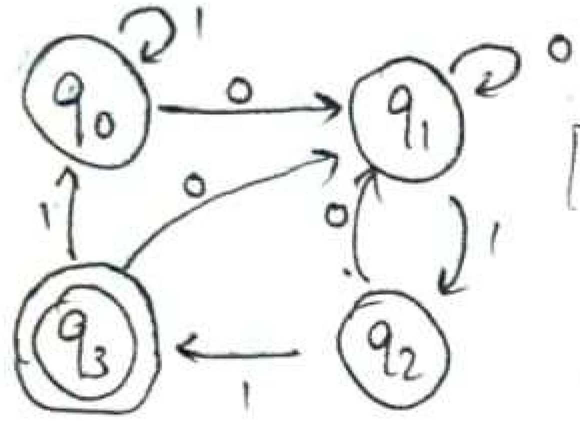
② $|y| > 0$

③ $\forall i \geq 0,\ xy^iz \in L$

Example:

① $\{w \mid w$ ends with $011\}$

If you recall, the corresponding DFA had 4 states.



[relabelled DFA]

Let $p = 4$.

So, if we take any $|w| \geq 4$, can use pigeonhole!

So say, $w = 10011$

$q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{1} q_3$

↳ notice repetition of q_1?

$w = xyz$, $x = 10$, $y = 0$, $z = 11$, satisfies conditions!

→ And, $xy^iz \rightarrow 10(0)^i11 \in L \;\forall i \geq 0$

② $L = 1011(01)^*11$

take $p = 7$, $w = xyz$, $x = 1011$, $z = 11$

y will be a +ve ~~multiple~~ no. of times '01'

So, xy^iz <u>will</u> belong.

③ $L = (01)^* = \{\varepsilon, 01, 0101, \dots\}$

$\boxed{p=2}$

$w = xyz$ — $x = \varepsilon$, $y = 01$, z ↳ some number of 01s, may be ε.

xy^iz <u>will</u> belong $\forall i \geq 0$.

eg. $w = 010101$, take $z = 0101$

$(01)^*0101 \in L$

④ L is a finite language.

Is it regular? → $\boxed{\text{Yes}}$

What will be the p then?

↳ p will be length of longest string $+1$

⑤ L has a DFA w/ n states. [$\Rightarrow L$ is regular]

Let $p = n$.

$\Rightarrow$ only consider strings of length $\geq n$.

So, we pass thru. $\geq n+1$ states for each such string, by pigeonhole, we <u>will</u> have a repetition.

Let $w = a_1 \dots a_n z$, [$a_i \in \Sigma$, $z \in \Sigma^*$]

$\Rightarrow \exists\, i, j$ such that,

$\hat{\delta}_D(q, a_i \dots a_j) = q$ [q is our repeated state]

$$q_0 \;\text{---}\; q \circlearrowleft^{a_i \dots a_j}$$

Let $x = a_1 \dots a_{i-1}$, $y = a_i \dots a_j$, $z_0 = a_{j+1} \dots a_n z$

$\therefore w = xyz_0 \Rightarrow xy^iz_0 \in L, \forall i \geq 0$.

as q is just repeated multiple times, just a loop!

⑥ $\{a^nb^n \mid n \geq 0\}$ is not regular!

$\Sigma = \{a, b\}$

↳ let's assume this is regular.

$\Rightarrow \exists\, p \geq 1$ such that p is pumping length.

Let $w = a^pb^p \rightarrow |w| = 2p \geq p$ & $w \in L$

$|xy| \leq p \Rightarrow xy$ has to substring of a^p

$\Rightarrow y$ is of all a's, say $y = a^l, 1 \leq l \leq p$

$\therefore xy^2z = a^pa^lb^p = a^{p+l}b^p$

$p + l \neq p \Rightarrow xy^2z \notin L$

$\Rightarrow \Leftarrow$, L is <u>not</u> regular.

Exercise:

Determine if $L = \{a^{n^2} \mid n \geq 0\}$ is regular or not using pumping lemma.